

YieldCore AI – Final Revision Brief for Replit

Objective

Transform YieldCore from a dashboard demo into a credible enterprise-grade AI farm operating system suitable for commercial farmers, investors, auditors, and enterprise customers.

Critical Fix #1 – Bio-Stack ROI

Current ROI values of 4000%–5000% are mathematically correct but commercially unrealistic. Target ROI range: 150%–600% for most farms. Exceptional cases may reach 800%–1200%. Increase Bio-Stack implementation costs and maintain realistic profit gains.

Critical Fix #2 – KPI Traceability

Every major KPI must include a 'View Calculation' drill-down. Show source data, assumptions, formulas, and expected outcomes. Users must understand exactly how Revenue Impact, Yield Increase, Water Savings, Carbon Credits, and Net Profit Increase are calculated.

Executive Summary Section

Add an AI Executive Summary at the top: - Treat Fall Armyworm - Irrigate Karoo House - Reorder MAP Fertiliser - Service Toyota Land Cruiser Display financial impact and urgency for each action. Show Total Actionable Value.

Farm Health Breakdown

Replace single health score with: Water Score Soil Score Pest Score Asset Score Inventory Score Overall Health Score

Carbon Credits

Display: Carbon Credits (tons) Estimated Carbon Revenue Carbon Price Assumptions Annual Financial Value

Energy Metrics

Clarify whether energy metrics are daily, monthly, or annual. Example: 931 kWh Today 340 MWh Annual Production R1.09M Annual Solar Savings

Inventory Improvements

Add: Days Remaining Reorder Threshold Projected Stockout Date Supplier Lead Time

AI Recommendations

Every recommendation should show: Confidence Estimated Financial Impact Risk Avoided Recommended Action

Asset Management

Show: Utilisation Downtime Trend Service Forecast Cost of Downtime Maintenance Savings

Benchmarking Improvements

Clearly identify benchmark sample size. Show whether benchmark is: Farm vs Farm Region vs Region Industry Average Demo Dataset

Investor Dashboard

Top summary metrics: Total Revenue Increase Average ROI Water Saved Carbon Reduction Energy Savings Procurement Savings Net Profit Increase

Enterprise Features

Add visibility for: Sensor Status Drone Flight Status Weather Station Status AI Model Update Time Data Freshness Indicators

Final Goal

The platform should feel like an AI Farm Operations Manager: Problem -> Detection -> Recommendation -> Action -> Outcome -> Financial Impact